

Assignment 5

1. File Head Display Script

Write a Bash script that accepts a filename as the first command-line argument:

- Check whether an argument is provided.
- Check whether the file exists using a file operator.
- If the file exists, display the first 5 lines using head.
- Otherwise, display an appropriate error message.

[filehead.sh]

```
#!/usr/bin/bash
if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi

if [ -f "$1" ]
then
    echo "First 5 lines of the file:"
    head -n 5 "$1"
else
    echo "Error: File does not exist."
fi
```

[Commands Executed]

```
mkdir Assignment_5_Theory
cd Assignment_5_Theory
nano filehead.sh
chmod +x filehead.sh
./filehead.sh sample.txt
```

2. File Type and Permission Checker

Write a Bash script that accepts a path as a command-line argument and determines whether it is a regular file, a directory, readable, writable, and executable. Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

[checkfile.sh]

261100690009 – Anshul A Salian

```
#!/usr/bin/bash
if [ -z "$1" ]
then
    echo "Error: No path provided."
    exit 1
fi

if [ -f "$1" ]
then
    echo "$1 is a regular file."
fi

if [ -d "$1" ]
then
    echo "$1 is a directory."
fi

if [ -r "$1" ]
then
    echo "$1 is readable."
else
    echo "$1 is not readable."
fi

if [ -w "$1" ]
then
    echo "$1 is writable."
else
    echo "$1 is not writable."
fi

if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi
```

[Commands Executed]

```
nano checkfile.sh
chmod +x checkfile.sh
```

261100690009 – Anshul A Salian

```
./checkfile.sh sample.txt
./checkfile.sh myFolder
```

3. Log File Search Script

Write a Bash script that accepts a log file as a command-line argument and searches for the words: failed, denied, unauthorized, and error.

[searchlog.sh]

```
#!/usr/bin/bash
if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -f "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi

grep -Ei "failed|denied|unauthorized|error" "$1"
```

[Commands Executed]

```
nano searchlog.sh
chmod +x searchlog.sh
./searchlog.sh security.log
```

4. Comprehensive File Validation & Preview Script

Write a Bash script that accepts a filename and checks whether the file exists, is a regular file, and is empty or non-empty. If it is non-empty, display the first 5 lines.

[filecheck.sh]

```
#!/usr/bin/bash
if [ -z "$1" ]
then
```

261100690009 – Anshul A Salian

```
    echo "Error: No filename provided."
    exit 1
fi

if [ -e "$1" ]
then
    echo "File exists."
else
    echo "File does not exist."
    exit 1
fi

if [ -f "$1" ]
then
    echo "It is a regular file."
else
    echo "It is not a regular file."
    exit 1
fi

if [ -s "$1" ]
then
    echo "File is non-empty."
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi
```

[Commands Executed]

```
nano filecheck.sh
chmod +x filecheck.sh
./filecheck.sh sample.txt
```

5. Interactive Log Investigation Script

Write a Bash script named `investigate.sh` that accepts a log file as a command-line argument. The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term and display the first 5 matching records along with the total count.

[investigate.sh]

261100690009 – Anshul A Salian

```
#!/usr/bin/bash
if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -e "$1" ]
then
    echo "Error: File does not exist."
    exit 1
fi

if [ ! -r "$1" ]
then
    echo "Error: File is not readable."
    exit 1
fi

read -p "Enter search term: " term
echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5
count=$(grep -ic "$term" "$1")
echo "Total matching records: $count"
```

[Commands Executed]

```
nano investigate.sh
chmod +x investigate.sh
./investigate.sh security.log
```